CS501 – Theory of Computation

Unit – I

- **Main**
  - Introduction of Automata Theory
  - Finite Automata
  - Moore and Mealy Machines
- **Detail**
  - Introduction of Automata Theory (examples of automata machines)
  - Finite Automata (language acceptor – translator)
  - Moore machines (properties)
  - Mealy machines (properties)
  - Composite machine
  - Conversion (Mealy to Moore – Moore to Mealy)

Unit – II

- **Main**
  - Types of Finite Automata
  - Regular Expressions
  - DFA Concepts
- **Detail**
  - Finite Automata (NDFA – DFA)
  - Conversion of NDFA to DFA
  - Minimization of automata machines
  - Regular expression
  - Arden's theorem
  - Operations (union – intersection – concatenation – closure)
  - 2 way DFA

Unit – III

- **Main**
  - Grammars
  - Context Free Grammar
  - Normal Forms
- **Detail**
  - Types of grammar
  - Context sensitive grammar
  - Context free grammar
  - Regular grammar
  - Derivation trees
  - Ambiguity in grammar
  - Simplification of context free grammar
  - Grammar ↔ Automata conversion

    - Chomsky hierarchy
    - Null & unit productions
    - Chomsky normal form
    - Greibach normal form

Unit – IV

- **Main**
    - Push Down Automata
    - PDA Types
    - Petrinet Model
- **Detail**
    - Example of PDA
    - Deterministic PDA
    - Non-deterministic PDA
    - PDA ↔ CFG conversion
    - CFG equivalent to PDA
    - Petrinet model

Unit – V

- **Main**
    - Turing Machine
    - Decidability
    - N-P Complete Problems
- **Detail**
    - Construction techniques
    - Universal Turing machine
    - Multitape – Multihead – Multidimensional TM
    - Recursively enumerable languages
    - Decidable – Undecidable languages
    - Halting problem
    - Post correspondence problem

CS502 – Database Management Systems

Unit – I

- **Main**
  - DBMS Concepts
  - Data Models
  - ER Model
- **Detail**
  - Database vs file system
  - Advantages of DBMS
  - Schemas – Instances
  - Data independence
  - DB languages – Interfaces
  - DBA functions
  - ER model (entities – attributes)
  - Generalization – Aggregation – Specialization
  - Object – Network – Relational models

Unit – II

- **Main**
  - Relational Model
  - SQL
  - Relational Algebra
- **Detail**
  - Domains – Tuples – Attributes – Relations
  - Keys – Integrity constraints
  - Referential integrity
  - SQL (DDL – DML)
  - Joins – Indexing – Triggers – Assertions
  - Relational algebra operations
  - Relational calculus

Unit – III

- **Main**
  - Normalization
  - Dependencies
  - Query Optimization
- **Detail**
  - Normal forms
  - Functional dependency
  - Decomposition
  - Dependency preservation
  - Lossless join

    - Multivalued dependency
    - Optimization techniques

Unit – IV

- **Main**
    - Transactions
    - Concurrency Control
    - Distributed Databases
- **Detail**
    - Serializability
    - Recovery techniques
    - Deadlock handling
    - Locking – Timestamp – Validation
    - Multiversion schemes
    - Data mining – Warehousing
    - OODBMS vs DBMS

Unit – V

- **Main**
    - Oracle / MySQL
    - SQL Programming
    - Database Security
- **Detail**
    - Architecture – Files – Memory
    - Tablespace – Segments – Extents
    - SQL joins – Subqueries
    - ANSI SQL
    - Cursors
    - Procedures – Functions – Triggers

CS503 (A) – Data Analytics

Unit – I

- **Main**
  - Statistics
  - Regression
  - ANOVA
- **Detail**
  - Descriptive statistics
  - Probability distributions
  - Inferential statistics
  - Hypothesis testing

Unit – II

- **Main**
  - Big Data
  - Hadoop
  - Analytics
- **Detail**
  - Four V's of Big Data
  - Big data analytics applications
  - Hadoop ecosystem
  - Cloud analytics
  - Predictive analytics

Unit – III

- **Main**
  - Big Data Processing
  - Data Integration
  - MapReduce
- **Detail**
  - Data extraction
  - Data transformation
  - Data subdivision
  - MapReduce preparation

Unit – IV

- **Main**
  - Hadoop MapReduce
  - HDFS
  - Execution Modes

- **Detail**
  - Job components
  - Distributed processing
  - Hadoop daemons
  - Local – Pseudo – Fully distributed

Unit – V

- **Main**
  - Big Data Tools
  - Pig
  - Hive
- **Detail**
  - Pig Latin – UDFs
  - Data processing operators
  - HiveQL
  - Oracle Big Data

CS503 (B) – Pattern Recognition

Unit – I

- **Main**
  - Pattern Recognition
  - Learning Models
  - Decision Concepts
- **Detail**
  - Datasets – Applications
  - Classification – Clustering
  - Supervised – Unsupervised learning

Unit – II

- **Main**
  - Classification Techniques
  - ML Models
  - Training Methods
- **Detail**
  - Decision tree
  - Naïve Bayes
  - SVM – Random forest
  - KNN
  - Normalization

Unit – III

- **Main**
  - Pattern Representation
  - Clustering
  - Validation
- **Detail**
  - K-means
  - Hierarchical clustering
  - Cluster validation

Unit – IV

- **Main**
  - Feature Selection
  - Optimization
  - Algorithms
- **Detail**
  - Feature extraction

    - Branch & bound
    - Sequential selection

Unit – V

- **Main**
    - Recent Advances
    - Soft Computing
    - Fuzzy Techniques
- **Detail**
    - Structural PR
    - Neuro-fuzzy
    - Density estimation
    - Nearest neighbor

CS503 (C) – Cyber Security

Unit – I

- **Main**
  - Cyber Crime
  - Threats
  - Attacks
- **Detail**
  - Cyber crime challenges
  - E-mail spoofing
  - Spamming
  - Internet time theft
  - Salami attack

Unit – II

- **Main**
  - Online Crimes
  - Network Attacks
  - Identity Theft
- **Detail**
  - Web jacking
  - Online frauds
  - Password sniffing
  - Session hijacking

Unit – III

- **Main**
  - Cyber Law
  - IT Act
  - Crime Control
- **Detail**
  - IT Act 2000
  - Cyber fraud
  - Defamation – Harassment
  - Monetary penalties

Unit – IV

- **Main**
  - Digital Evidence
  - Legal Aspects
  - Electronic Records

- **Detail**
  - Indian Evidence Act 1872
  - Digital signatures
  - Electronic agreements

Unit – V

- **Main**
  - Cyber Crime Tools
  - Malware
  - Attacks
- **Detail**
  - Proxy servers – Anonymizers
  - Virus – Worms – Trojans
  - DoS – DDoS
  - Phishing

CS504 (A) - Internet and Web Technology

Unit - I

- **Main**
  - WWW Concepts
  - Web Design
  - Website Planning
- **Detail**
  - Internet and WWW
  - HTTP protocol (request - response)
  - Web browsers - Web servers
  - Web 2.0 features
  - Effective web design concepts
  - Web design issues (browser - bandwidth - cache)
  - Page layout - linking
  - User centric design
  - Sitemap - planning - publishing website
  - Designing effective navigation

Unit - II

- **Main**
  - HTML
  - Web Structure
  - HTML5
- **Detail**
  - Basics of HTML
  - Formatting - fonts - comments
  - Colors - hyperlinks - lists - tables
  - Images - forms
  - XHTML
  - Meta tags - character entities
  - Frames - frame sets
  - Browser architecture
  - Website structure
  - HTML5 features

Unit - III

- **Main**
  - CSS

    - JavaScript
    - DHTML
- **Detail**
    - CSS need – syntax – structure
    - Background – colors – text – fonts
    - Borders – boxes – margin – padding
    - Positioning using CSS
    - CSS2 – CSS3
    - JavaScript basics (variables – functions – conditions – loops)
    - Pop up boxes
    - JavaScript objects – DOM
    - Form validation
    - DHTML (HTML – CSS – JavaScript)

Unit – IV

- **Main**
    - XML
    - XSL / XSLT
    - PHP
- **Detail**
    - XML introduction – uses
    - XML components
    - DTD – schemas
    - XML with applications
    - Transforming XML using XSL – XSLT
    - PHP syntax
    - Decision – looping
    - PHP with HTML
    - Arrays – functions – strings
    - Cookies – sessions
    - Object oriented PHP

Unit – V

- **Main**
    - PHP and MySQL
    - Database Operations
    - PHP Tools
- **Detail**
    - Server connection
    - Create – select – delete database
    - Tables – insert – alter – delete

- Queries
- PHP myadmin
- Database bugs

CS504 (B) – Object Oriented Programming

Unit – I

- **Main**
  - Programming Basics
  - Data Types
  - Operators
- **Detail**
  - Character set
  - Constants – variables
  - Keywords – identifiers
  - Type declaration
  - Arithmetic – relational – logical operators
  - Type conversion
  - Operator hierarchy

Unit – II

- **Main**
  - Control Structures
  - Loops
  - Jump Statements
- **Detail**
  - If – if else
  - Logical operators
  - While – for – do while
  - Nested loops
  - Break – continue
  - Switch – case
  - Goto – exit

Unit – III

- **Main**
  - Arrays
  - Strings
  - Memory Layout
- **Detail**
  - 1D – 2D arrays
  - Array initialization
  - Bound checking

    - Memory map
    - Strings
    - String functions (strlen – strcpy – strcat – strcmp)
    - 2D array of characters

Unit – IV

- **Main**
    - Structures
    - Preprocessor
    - Union
- **Detail**
    - Structure declaration – access
    - Array of structures
    - Uses of structure
    - Macro expansion
    - File inclusion
    - Conditional directives
    - Union declaration
    - Union of structures
    - User defined data types

Unit – V

- **Main**
    - OOP Concepts
    - Program Structure
    - Applications
- **Detail**
    - Object – class
    - Data abstraction – encapsulation
    - Inheritance – polymorphism
    - Static – dynamic binding
    - Message passing
    - Advantages – disadvantages
    - Applications of OOP
    - Program creation – compilation – linking

CS504 (C) – Introduction to Database Management Systems

Unit – I

- **Main**
  - DBMS Concepts
  - Database Applications
  - DBMS vs RDBMS
- **Detail**
  - Database significance
  - Data independence
  - Data modeling
  - Entities – attributes – relationships
  - Advantages – disadvantages of DBMS

Unit – II

- **Main**
  - Database Models
  - ER Model
  - Associative Model
- **Detail**
  - Relational model
  - Hierarchical model
  - Network model
  - Object oriented model
  - ER diagram notation
  - Relationship types

Unit – III

- **Main**
  - SQL DDL
  - Constraints
  - Index
- **Detail**
  - Create – drop – alter table
  - Primary key – foreign key
  - Truncate table
  - Cursor

Unit – IV

- **Main**
  - SQL DML
  - Data Modification
  - Merge
- **Detail**
  - Insert statement
  - Delete with condition
  - Update statement
  - Merge statement

Unit – V

- **Main**
  - SQL SELECT
  - Queries
  - ANSI SQL
- **Detail**
  - Joins (equi – non equi – self – outer)
  - Special operators (like – any – all – exists – in)
  - Hierarchical queries
  - Inline queries
  - Flashback queries
  - Anonymous block
  - Branching – looping constructs